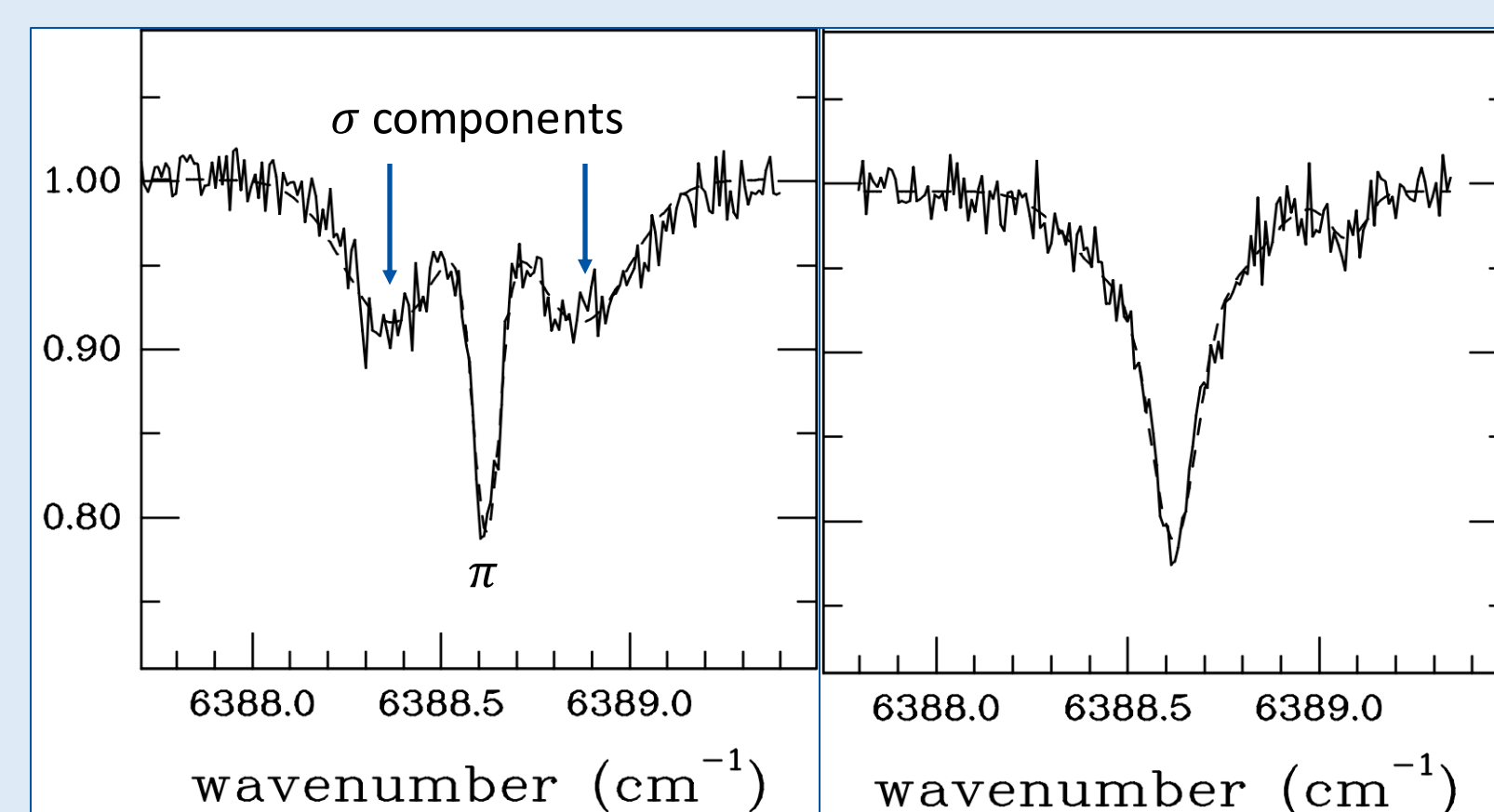


Abstract

We investigate how solar magnetic activity influences individual spectral lines in high-resolution near-infrared solar spectra. In particular, we aim to assess whether similar infrared observations in Sun-like stars can help mitigate stellar noise in radial velocity measurements used to detect exoplanets. Using the SoCalSoCal solar tracker and the PARVI high-resolution ($R \approx 80,000$) spectrograph at Palomar Observatory, we obtained and analyzed daily solar spectra over a few months, focusing on a strongly Zeeman-sensitive Fe I absorption line near 1565 nm. After correcting the data for micro-telluric absorption around this line, we measured changes in its equivalent width and applied periodogram analysis to search for activity-related signals. Once instrumental and atmospheric effects were accounted for, we found evidence of activity-driven line-shape variations, particularly at the solar rotation period.

Background & Motivation

- We can find exoplanets by measuring the tiny Doppler “wobble” of a star’s light (radial-velocity method), but Earth-size planets produce extremely small signals.
- Magnetic regions and spots on the star’s surface change the shapes of spectral lines (“stellar jitter”), which can mimic or hide the planet signal.
- Some near-infrared (NIR) iron lines are very sensitive to magnetic fields: they show Zeeman splitting, where one line splits into several components in a strong field (see image below).
- The Sun is the ideal test case because we can see its surface while also measuring its NIR spectrum as if it were a single distant star.
- By tracking a Zeeman-sensitive Fe line in daily solar spectra, we test whether NIR line-shape changes can serve as activity diagnostics to improve RV planet searches around Sun-like stars.



Methods

- Collected daily disk-integrated NIR solar spectra using SoCalSoCal + PARVI
- Selected a continuous 63-day window for stable periodogram analysis.
- Corrected atmospheric H₂O/HDO absorption using Stellar-GFIT.
- Normalized spectra and measured equivalent width (EW) for Fe I, Ni I, and H₂O lines.
- Computed Lomb–Scargle periodograms to search for rotation-linked signals.

Results & Discussion

- The Fe I 6388.6 cm⁻¹ (1565 nm) line shows clear periodic structure linked to solar magnetic activity.
- A strong peak near ~28 days, matching the solar rotation period.
- Secondary peaks at ~18–19 days and ~31 days, likely due to differential rotation or sampling.
- The telluric H₂O line does not show this pattern → confirms successful atmospheric correction.
- Across Fe and Ni lines, rotation-linked variability is consistent, reflecting sensitivity to magnetic topology.

Conclusion

- Analyzed the line-by-line behavior of Zeeman-sensitive near-infrared solar lines in daily SoCalSoCal + PARVI spectra.
- Showed that time-series analysis is strongly affected by instrumental systematics, atmospheric (“telluric”) absorption, and data gaps, requiring extensive data cleaning and telluric modeling.
- Found clear signatures of solar magnetic activity in the Fe I line at 6388.6 cm⁻¹ (1565 nm), indicating that NIR line-shape changes can serve as magnetic-activity diagnostics for future radial-velocity exoplanet searches.

Acknowledgements

I thank G. Vasisht, S. Halverson, and K. Sung for their mentorship and guidance, and the JPL-Caltech SURF program and ROSES-EPRV Infrared Sun-as-a-Star grant for supporting this work. I also thank the staff at Palomar Observatory for their assistance with instrumentation.

